

SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE
SUBJECT - Data Analysis with R/SAS/SPSS

Aim - Performing one-sample t-tests using t.test() (R).

Output :

The first screenshot shows the initial data loading and summary statistics for the 'penguins.csv' dataset. The second screenshot shows the results of three one-sample t-tests performed on the 'culmen_length' variable.

```
> rm(list = ls())
> data <- read.csv("penguins.csv")
> head(data)
  species      island culmen_length culmen_depth flipper_length body_mass sex
1 Adelie Torgersen   39.1         18.7         181        3750    MALE
2 Adelie Torgersen   39.5         17.4         186        3800    FEMALE
3 Adelie Torgersen   40.3         18.0         195        3250    FEMALE
4 Adelie Torgersen    NA          NA          NA          NA    <NA>
5 Adelie Torgersen   36.7         19.3         193        3450    FEMALE
6 Adelie Torgersen   39.3         20.6         190        3650    MALE
> str(data)
'data.frame':   344 obs. of  7 variables:
 $ species      : chr  "Adelie" "Adelie" "Adelie" "Adelie" ...
 $ island       : chr  "Torgersen" "Torgersen" "Torgersen" "Torgersen" ...
 $ culmen_length: num  39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ culmen_depth : num  18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length: int   181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass    : int   3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex         : chr   "MALE" "FEMALE" "FEMALE" NA ...
> summary(data$culmen_length)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
 32.10  39.23  44.45  43.92  48.50  59.60     2
> summary(data$culmen_depth)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
 13.10  15.60  17.30  17.15  18.70  21.50     2
> summary(data$flipper_length)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
 172.0  190.0  197.0  200.9  213.0  231.0     2
> summary(data$body_mass)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
 2700   3550   4050   4202   4750   6300     2
> t.test(data$culmen_length, mu = 0)

One Sample t-test

data: data$culmen_length
t = 148.78, df = 341, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 43.34125 44.50261
sample estimates:
mean of x
43.92193
```

```
> t.test(data$culmen_length, mu = 50, conf.level = 0.95)

One Sample t-test

data: data$culmen_length
t = -20.588, df = 341, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 50
95 percent confidence interval:
 43.34125 44.50261
sample estimates:
mean of x
43.92193
> t.test(data$culmen_length, mu = 50, alternative = "greater")

One Sample t-test

data: data$culmen_length
t = -20.588, df = 341, p-value = 1
alternative hypothesis: true mean is greater than 50
95 percent confidence interval:
 43.43501      Inf
sample estimates:
mean of x
43.92193
> t.test(data$culmen_length, mu = 50, alternative = "less")

One Sample t-test

data: data$culmen_length
t = -20.588, df = 341, p-value < 2.2e-16
alternative hypothesis: true mean is less than 50
95 percent confidence interval:
 43.34125 44.50261
sample estimates:
mean of x
43.92193
```

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The image displays two screenshots of the RStudio interface, showing the results of t-tests performed on data from the 'penguins' dataset.

Top Screenshot: The console shows the results of a t-test for 'data\$culmen_depth' with a null hypothesis of $\mu = 15$. The test is a one-sample t-test. The results are as follows:

- One Sample t-test
- data: data\$culmen_depth
- t = 20.145, df = 341, p-value < 2.2e-16
- alternative hypothesis: true mean is not equal to 15
- 95 percent confidence interval: 16.94113 17.36121
- sample estimates: mean of x 17.15117

The test is repeated for two alternative hypotheses: 'greater' and 'less'.

Bottom Screenshot: The console shows the results of a t-test for 'data\$flipper_length' with a null hypothesis of $\mu = 200$. The test is a one-sample t-test. The results are as follows:

- One Sample t-test
- data: data\$flipper_length
- t = 1.2036, df = 341, p-value = 0.2296
- alternative hypothesis: true mean is not equal to 200
- 95 percent confidence interval: 199.4196 202.4108
- sample estimates: mean of x 200.9152

The test is repeated for two alternative hypotheses: 'greater' and 'less'.

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The screenshot displays the RStudio interface. The console on the left shows the execution of three t-test commands on the 'data\$body_mass' variable. The first command tests for equality with a null hypothesis of 4000. The second tests for a greater mean, and the third tests for a less mean. The Environment pane on the right shows the 'data' object with 344 observations and 7 variables.

```
R - R4.5.2 - ~/R
> t.test(data$body_mass, mu = 4000, conf.level = 0.95)

One Sample t-test

data: data$body_mass
t = 4.6525, df = 341, p-value = 4.7e-06
alternative hypothesis: true mean is not equal to 4000
95 percent confidence interval:
 416.458 4287.050
sample estimates:
mean of x
 4201.754

> t.test(data$body_mass, mu = 4000, alternative = "greater")

One Sample t-test

data: data$body_mass
t = 4.6525, df = 341, p-value = 2.35e-06
alternative hypothesis: true mean is greater than 4000
95 percent confidence interval:
 4130.231      Inf
sample estimates:
mean of x
 4201.754

> t.test(data$body_mass, mu = 4000, alternative = "less")

One Sample t-test

data: data$body_mass
t = 4.6525, df = 341, p-value = 1
alternative hypothesis: true mean is less than 4000
95 percent confidence interval:
      -Inf 4273.277
sample estimates:
mean of x
 4201.754

>
```

Environment

Object	Class	Attributes
data	data.frame	344 obs. of 7 variables

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